Pre S3 L01-02 More about Factorization Quiz
(ANSWER)

/42

Unit Education

Name: ____________________

Date: ____________________

1. Factorize $x^2 - 6x - 7$

   A. $(x+1)(x+7)$

   B. $(x+1)(x-7)$

   C. $(x-1)(x+7)$

   D. $(x-1)(x-7)$

The answer is B.

B

2. Factorize $49p^2 + 9q^2 - 42pq$

   A. $(7p+3q)(7p-3q)$

   B. $(7p+3q)^2$

   C. $(7p-3q)^2$

   D. $(49p+9q)^2$

$$\begin{aligned} 49p^2 + 9q^2 - 42pq &= 49p^2 - 42pq + 9q^2 \\ &= (7p)^2 - 2(7p)(3q) + (3q)^2 \\ &= \underline{\underline{(7p-3q)^2}} \end{aligned}$$

C

3. Which of the following do(es) not have $x+2$ as a factor?

   I. $x^2 + 4$

   II. $x^2 - 4$

   III. $(x-3)^2 - 25$

   A. I only

   B. II only

   C. I and III only

   D. II and III only

II: $x^2 - 4 = (x+2)(x-2)$

III: $(x-3)^2 - 25 = (x-3)^2 - 5^2$
$= (x-3+5)(x-3-5)$
$= (x+2)(x-8)$

$\therefore$ Only $x^2 + 4$ (i.e. I) does not have $x+2$ as a factor.

A

4. Factorize $x^2 - 8xy + 15y^2 - 5x + 15y$

   A. $(x-3y)(x-5y-5)$

   B. $(x-5y)(x-3y-5)$

   C. $(x+3y)(x-5y-5)$

   D. $(x+5y)(x-3y-5)$

$$\begin{aligned} x^2 - 8xy + 15y^2 - 5x + 15y &= (x^2 - 8xy + 15y^2) - (5x - 15y) \\ &= (x-3y)(x-5y) - 5(x-3y) \\ &= \underline{\underline{(x-3y)(x-5y-5)}} \end{aligned}$$

A

5. Factorize $x^2(x+y) - y^2(y+x)$

   A. $(x+y)^2(x-y)$

   B. $(x+y)(x^2+xy+y^2)$

   C. $(x-y)^2(x+y)$

   D. $(x-y)(x^2-xy+y^2)$

$$\begin{aligned} & x^2(x+y) - y^2(y+x) \\ &= x^2(x+y) - y^2(x+y) \\ &= (x+y)(x^2 - y^2) \\ &= (x+y)(x+y)(x-y) \\ &= \underline{\underline{(x+y)^2(x-y)}} \end{aligned}$$

A

6. Factorize $y^2 + 4y - 12$

   A. $(y+2)(y-6)$

   B. $(y+3)(y-4)$

   C. $(y-2)(y+6)$

   D. $(y-3)(y+4)$

The answer is C.

C

7. Factorize $n^2 + 12n + 35$

   A. $(n+5)(n+7)$

   B. $(n+5)(n-7)$

   C. $(n-5)(n+7)$

   D. $(n-5)(n-7)$

The answer is A.

A

8. Factorize

(a) $x^2 + 8x + 7$ (2 marks)

(b) $y^2 - 11y - 26$ (2 marks)

(a) $x^2 + 8x + 7 = \underline{\underline{(x+1)(x+7)}}$ 1M+1A

(b) $y^2 - 11y - 26 = \underline{\underline{(y+2)(y-13)}}$ 1M+1A

9. (a) Factorize $x^2 - 2xy - 35y^2$ (2 marks)

(b) Hence, factorize $x^2 - 2xy - 35y^2 - 7x + 49y$ (3 marks)

$(a)\quad x^2 - 2xy - 35y^2$

$= (x - 7y)(x + 5y) \qquad 1M + 1A$

$(b)\quad x^2 - 2xy - 35y^2 - 7x + 49y$

$= (x - 7y)(x + 5y) - 7(x - 7y) \qquad 1M$

$= (x - 7y)(x + 5y - 7) \qquad 1M + 1A$

10. Factorize

(a) $240uv - 72u^2 - 200v^2$ (3 marks)

(b) $63xy^2 + 28xz^2 + 84xyz$ (2 marks)

(a) $240uv - 72u^2 - 200v^2 = -8(-30uv + 9u^2 + 25v^2)$ 1M

$= -8(9u^2 - 30uv + 25v^2)$

$= -8[(3u)^2 - 2(3u)(5v) + (5v)^2]$ 1M

$= \underline{\underline{-8(3u - 5v)^2}}$ 1A

(b) $63xy^2 + 28xz^2 + 84xyz = 7x(9y^2 + 4z^2 + 12yz)$

$= 7x(9y^2 + 12yz + 4z^2)$

$= 7x[(3y)^2 + 2(3y)(2z) + (2z)^2]$ 1M

$= \underline{\underline{7x(3y + 2z)^2}}$ 1A

*11. Factorize $a^4 - a^2 - 2a^2b^2 - 2ab + b^4 - b^2$. (7 marks)

$a^4 - a^2 - 2a^2b^2 - 2ab + b^4 - b^2$

$= a^4 - 2a^2b^2 + b^4 - a^2 - 2ab - b^2$

$= (a^2)^2 - 2a^2b^2 + (b^2)^2 - (a^2 + 2ab + b^2)$ 1M+1M

$= (a^2 - b^2)^2 - (a + b)^2$ 1A+1A

$= [(a + b)(a - b)]^2 - (a + b)^2$ 1A

$= (a + b)^2(a - b)^2 - (a + b)^2$

$= (a + b)^2[(a - b)^2 - 1]$ 1M

$= (a + b)^2[(a - b)^2 - 1^2]$

$= (a + b)^2[(a - b) + 1][(a - b) - 1]$

$= \underline{\underline{(a + b)^2(a - b + 1)(a - b - 1)}}$ 1A